

# Point of Sale Fraud Detection Methods via Machine Learning

Emre Begen

*Data Science and Innovation  
Ata Technology Platforms  
İstanbul, Türkiye  
emrebe@atp.com.tr*

İsmail Utku Sayan

*Data Science and Innovation  
Ata Technology Platforms  
İstanbul, Türkiye  
utkus@atp.com.tr*

Ahmet Tuğrul Bayrak

*Data Science and Innovation  
Ata Technology Platforms  
İstanbul, Türkiye  
tugrulb@atp.com.tr*

Olca Taner Yıldız

*Computer Engineering  
Özyeğin University  
İstanbul, Türkiye  
olcay.yildiz@ozyegin.edu.tr*

**Abstract**—Restaurant cash registers frequently experience fraudulent transactions, leading to substantial financial losses for operators. Despite several methods aimed at preventing fraud at the cash register, addressing this issue remains an ongoing concern. In this study, machine learning methods are used to detect fraudulent transactions at the cash register in fast-food restaurants. By using POS logs, transactions in restaurants are recorded and these logs are analyzed to detect fraudulent transactions on an unbalanced dataset. Random forest, XGBoost and LGBM algorithms are used in the study and different resampling techniques (ADASYN etc.) are applied to improve the performance of these algorithms. In addition, it is aimed to find the best parameters with the randomized search method. In conclusion, this study offers a solution for detecting fraudulent transactions at the cash register in fast-food restaurants. The results of the study are promising in its current state.

**Index Terms**—point of sale fraud detection, unbalanced data, resampling, xgboost, lgbm

## I. INTRODUCTION

Fast food products are among the places of food consumption preferred by many consumers due to their advantages, such as short preparation times, reasonable prices and comprehensive menu options. However, checkout fraud is a common problem at many fast-food restaurants. Such fraudulent transactions are carried out without customers noticing, especially during peak hours. For example, tricks such as entering the prices of the products sold at the cash register on purpose, underestimating the amount of money given by the customers, or collecting unnecessary additional fees can be done. Detecting such fraudulent transactions is extremely important to increase customer satisfaction and protect the reputation of the business. In this context, it is possible to detect such fraudulent transactions with machine learning methods using point of sale (POS) logs.

Fraud detection is a critical concept for businesses. In almost every sector, malicious uses and abuse events can be seen. Such suspicious transactions are carried out both digitally and manually. However, fraudulent transactions can be made by users as well as by business employees. There is a need to control the sales and the system, especially in restaurants that provide fast service in the service sector. In the mentioned sector, the order may be shown as a different order, the cash may not be put in the register, and the receipt may not be

cut after the sale. Currently, these transactions are performed by manually checking and comparing the camera images of thousands of suspicious transactions and sales data. Although it takes much time to analyze these data manually, especially in chain brands with a large number of branches, it is not possible to examine all of the data.

Restaurant fraud prevention projects focus on detecting suspicious transactions and preventing loss/illegal transactions by processing sales data made through fast service in the retail industry, package service and many similar online channels. According to research, pos fraud transactions cause businesses to lose about 8 billion dollars a year [1], [2]

In this study, it is aimed to detect fraudulent transactions at the cash register in a fast-food restaurant using pos logs. These logs record all transactions performed in the cash registers of the business. By analyzing these logs using machine learning methods, it is targeted to detect fraudulent transactions. This study offers a solution for food consumption operators such as fast food restaurants to help them maintain customer satisfaction and reputation.

With the developed project, datasets collected from pos sales data and user movements, risky transactions that will be processed by machine learning and that may be subject to audit can be reported automatically. Thus, it is aimed to analyze and report the data determined as relatable from a common platform. Thanks to this development, the relevant teams will be able to conduct their audits. In addition, this application aims to provide fast and up-to-date data in reports. However, since the number of transactions with actual fraud is deficient, the data set in the study shows an unbalanced distribution. For this reason, different oversampling methods are used and compared.

## II. LITERATURE REVIEW

Alabi et al. try to detect fraudulent transactions by developing a logistic regression model on pos data [3]. Similarly, [4] Hines et al. have implemented random forest and support vector machines (SVM) algorithms for in-restaurant fraud detection for small businesses. On the other hand, Collins compared rule-based methods to machine learning methods for fraud detection [5]. Long et al. detected fraud using local outlier factor (LOF) and K-means unsupervised learning methods [6].

Xing et al. have also implemented a hidden Markov model (HMM) on log strings in the vault [7]. However, Gabbur et al. have constructed a classification model by processing video footage [8]. Yoseph et al. tested and concluded clustering methods for outlier detection in the POS area. K-means, FCM-Fuzzy C Means are used [9].

Apart from these, in some studies, they detected fraud by generating artificial data with the retsim simulator [10], [11].

### III. METHOD

#### A. Data and Preprocessing Steps

In this study, it is aimed to detect fraud transactions from all the transactions executed on POS devices. A sample of data is selected. For this, 166 of the actual fraud cases are labelled as positive. However, 501894 amount of transaction data that are not fraudulent are marked as negative. Outline values in these log data are scored using the z-score. After these arrangements and the merging of the log data, the features seen in Table I are obtained.

TABLE I: Features of the Models

| Feature                 | Explanation                                              |
|-------------------------|----------------------------------------------------------|
| is_between_0912         | One-hot values based on the transaction hour             |
| is_between_1215         |                                                          |
| is_between_1518         |                                                          |
| is_between_1821         |                                                          |
| is_monday               | One-hot values based on the transaction day              |
| is_tuesday              |                                                          |
| is_wednesday            |                                                          |
| is_thursday             |                                                          |
| is_friday               |                                                          |
| is_weekend              | If the transaction day is on the weekend                 |
| from_expensive_to_cheap | If the deleted items more expensive than the added items |
| number_of_steps         | Total number of steps in a transaction                   |
| number_of_deleted_items | Total number of deleted items in a transaction           |
| total_amount            | Total amount of the transaction                          |

#### B. Oversampling Methods

After cleaning and analyzing the data containing suspicious transactions, the modelling process starts. Since the mentioned data is an unbalanced dataset, balancing the data before modelling will allow us to obtain much better classification results. The methods that can be applied in the classification of unbalanced data are as follows;

#### C. Class Weights

This approach increases the weight of the minority class in the model training process and enables the model to take this class into account more. This is mainly applied when the target variable is unstable. This method assumes that all instances are of equal importance within the same class. This assumption may not always be correct. For example, specific samples may contain more information than others, or some may be noisy or outliers. In these cases, a more complex sampling strategy or application of cost-sensitive methods may be more appropriate to obtain better results. Moreover, these situations

may require the use of different techniques in various classification problems. Therefore, the proper technique depends on the particular situation and dataset, and finding the best solution for each case often takes a trial and error process.

#### D. Resampling Techniques

This method is applied to oversample the instances of the minority class or to undersample the instances of the majority class.

- One of the oversampling methods is Synthetic Minority Over-sampling Technique (SMOTE). This method creates synthetic instances from the minority class. Adaptive Synthetic Sampling (ADASYN) follows a similar approach and focuses more on samples at the border of the minority class.
- Undersampling reduces instances of the majority class randomly or with a specific strategy. However, this method may result in the loss of important information from the majority class.

#### E. Cost-Sensitive Classification

This method stabilizes the algorithm by increasing the cost of misclassifying minority-class samples.

#### F. Ensemble Methods

There are several ensemble methods for the classification problem on unbalanced datasets. Balanced Random Forest and Easy Ensemble are examples of these methods.

1) *Deep Learning Methods*: Deep learning methods such as Autoencoders and Generative Adversarial Networks (GANs) can be used to generate new samples from the minority class.

#### G. Evaluation Metrics

It is critical to use metrics that are more appropriate for unstable data sets, such as precision, recall, F1-Score, and AUC-ROC. These metrics more accurately measure how well the model predicts the minority class.

In this project, resampling techniques are used to classify suspicious data better. RandomForest, XGBoost and LGBM algorithms are tried to achieve the best results by applying RandomizedSearch together with the following resampling techniques, respectively.

#### H. Adaptive Synthetic (ADASYN)

It is an improvement on the SMOTE algorithm. Both algorithms generate minority-class synthetic samples, but ADASYN takes into account the density of minority-class samples at class boundaries and generates more synthetic samples in these areas. This helps the model better learn the decision regions at the classroom boundaries.

The ADASYN algorithm follows the following steps: 1. Finds k-nearest neighbours for each minority class instance. 2. For each sample of the minority class, using the number of k-nearest neighbours belonging to the majority class, determine how difficult this sample is at the class boundaries. This measurement is used to determine the weight of the minority class sample. 3. Creates more synthetic samples for more complex (i.e., class boundary) samples. As a result, the ADASYN

algorithm better preserves the distribution of minority class samples and their relationship with the majority class and creates a more balanced learning algorithm. However, like other synthetic sampling techniques, such as SMOTE, the ADASYN algorithm can potentially increase noise and cause overfitting problems. This is because synthetic samples run the risk of not fully representing the actual infrastructure of the minority class.

### I. SMOTE

It is used to balance unbalanced class distributions by creating synthetic samples from the minority class. This algorithm follows the following steps: 1. A sample from the minority class is selected. This selection is made randomly or depending on your sampling strategy. 2. The  $k$  nearest neighbours of the selected sample are determined. Usually,  $k=5$  is chosen. 3. One of the  $k$  nearest neighbours is chosen randomly. 4. A new synthetic sample is created using linear interpolation between the selected minority class sample and the randomly selected nearest neighbour. SMOTE has several advantages over ROS (Random Over Sampling), which is a random sampling technique. In ROS, it relies on simply duplicating existing minority class instances, which increases the risk of model overfitting because it uses exactly the same instances over and over. On the other hand, synthetic samples are created in SMOTE, so the model improves its ability to generalize to new samples. However, as this method risks not fully representing the actual infrastructure of the minority class, it can potentially increase noise and lead to overfitting problems. Therefore, it is essential to always check the performance of the model after applying SMOTE and make parameter adjustments as needed.

### J. Random Under-Sampling (RUS)

It is a sampling technique that works by randomly extracting samples of the majority class from unbalanced data sets. This aims to reduce the imbalance between the number of instances of the minority class and the number of instances of the majority class. This technique can be beneficial in situations where the data set is extensive and where a considerable proportion of samples of the majority class repeat information. However, RUS has a significant disadvantage: samples that are excluded from the majority class lose information, which can negatively affect the overall performance of the model because the samples that are removed can contain potentially helpful information. The Random Under-Sampling technique is often used as a preprocessing step to reduce the size of the dataset and shorten the processing time. However, the use of Random Under-Sampling may reduce the model's ability to generalize accurately to the majority class because a significant fraction of the samples of the majority class is omitted. Therefore, caution should be exercised when using this technique and its combination with other techniques (for example, oversampling techniques) should be considered.

1) *Tomek's Links*: This method is used to identify links between a minority class instance and a majority class instance. Precisely, a minority class instance and a majority class instance form a Tomek Link if those two instances are nearest neighbours and not closer to another instance. This technique is often used to extract samples from the majority class. Thus, the boundaries between classes are made more clear. This is especially useful when instances of the majority class are very close to instances of the minority class. This makes the decision boundaries more transparent and potentially improves classification performance. Tomek's Links technique can be used as an undersampling technique to alleviate class imbalance problems. However, the computational cost of this technique can be high, especially with extensive datasets. Because in order to find the nearest neighbour of each sample, it is necessary to calculate the distance from all other samples. Tomek's Links are often used in conjunction with other techniques. For example, it can be combined with a technique such as SMOTE to create synthetic samples from a minority class. Then Tomek's Links can be used to extract instances from the majority class. This can help multiply instances of the minority class and reduce instances of the majority class, thereby easing the overall class imbalance.

2) *SMOTEENN*: It refers to a combination of SMOTE and Edited Nearest Neighbors (ENN) techniques. This hybrid method is often used to improve classification performance on unbalanced datasets. Using the two techniques together usually results in a dataset with smoother class boundaries, leading to better classification performance overall.

First, the SMOTE method is used to create more instances in the minority class. This is accomplished by creating synthetic instances between instances of the minority class. These generated synthetic samples help the classification algorithm to generalize this class better by extending the feature distribution of the minority class. Next, the ENN method is used. This is an undersampling technique to extract samples from the majority class. ENN examines a sample's nearest neighbours and infers that sample if the majority of a sample does not belong to the same class. This process helps to make the boundaries between classes clearer and generally improves classification performance. The combination of SMOTEENN techniques can be quite effective in mitigating class imbalance problems and generally improving classification performance. However, these techniques can be computationally expensive, especially on vast datasets.

## IV. EXPERIMENTS

A balanced dataset is obtained by using the analyzed and cleaned dataset and the resampling methods described above. Then the data is splitted as train and test sets with the percent 80% and 20%. Afterwards, it is targeted to reach the best parameters that give the best results in the XGBoost, LGBM and Random Forest classification algorithms and the RandomizedSearchCV hyperparameter tuning algorithm. Below are the F1 scores of the combinations of classification algorithms

TABLE II: MODEL PARAMETERS

| Sampling Methods | Models                                                                       |                                                      |                                                      |
|------------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
|                  | Random Forest                                                                | XGBoost                                              | LGBM                                                 |
| ADASYN           | n_estimators: 75, min_samples_split: 5, min_samples_leaf: 2, max_depth: 14   | n_estimators: 500, max_depth: 7, learning_rate: 0.1  | n_estimators: 500, max_depth: 7, learning_rate: 0.1  |
| SMOTE            | n_estimators: 75, min_samples_split: 5, min_samples_leaf: 2, max_depth: 14   | n_estimators: 500, max_depth: 7, learning_rate: 0.1  | n_estimators: 500, max_depth: 7, learning_rate: 0.1  |
| RUS              | n_estimators: 225, min_samples_split: 10, min_samples_leaf: 2, max_depth: 12 | n_estimators: 100, max_depth: 7, learning_rate: 0.01 | n_estimators: 100, max_depth: 7, learning_rate: 0.01 |
| ROS              | n_estimators: 75, min_samples_split: 5, min_samples_leaf: 2, max_depth: 14   | n_estimators: 500, max_depth: 7, learning_rate: 0.1  | n_estimators: 500, max_depth: 7, learning_rate: 0.1  |
| TL               | n_estimators: 75, min_samples_split: 5, min_samples_leaf: 2, max_depth: 14   | n_estimators: 500, max_depth: 7, learning_rate: 0.1  | n_estimators: 500, max_depth: 5, learning_rate: 0.1  |
| SMNN             | n_estimators: 75, min_samples_split: 5, min_samples_leaf: 2, max_depth: 14   | n_estimators: 500, max_depth: 7, learning_rate: 0.1  | n_estimators: 500, max_depth: 5, learning_rate: 0.1  |

and resampling methods created with RandomizedSearchCV. Models and the parameters applied can be seen in Table II.

Table III shows the models and their results. According to the results, it is seen that the models trained with Tomek's Links are more successful. In addition, the most successful model on a model basis is the XGBoost.

TABLE III: MODEL RESULTS

| Sampling Methods | Models        |         |         |
|------------------|---------------|---------|---------|
|                  | Random Forest | XGBoost | LGBM    |
| ADASYN           | 0.81944       | 0.92380 | 0.91725 |
| SMOTE            | 0.91319       | 0.92544 | 0.92316 |
| RUS              | 0.90534       | 0.90410 | 0.89829 |
| ROS              | 0.91429       | 0.91588 | 0.90954 |
| TL               | 0.92774       | 0.93228 | 0.92933 |
| SMOTEENN         | 0.91130       | 0.91959 | 0.91717 |

## V. RESULTS

Fraud transactions made at restaurant cash registers are common. For this reason, operators lose a significant amount

of money. Although there are various methods to prevent fraudulent transactions at the cash desk, this is still an issue that continues to be considered. In the study, pos logs in a fast-food chain are used. In addition to thousands of transactions, machine learning models have been established in which cases that are found to be genuinely fraudulent are also used. State-of-the-art oversampling methods have also been used, as the number of cases that are certain to be fraudulent is deficient. Performance comparisons are made for all models as a result of applying oversampling methods to different models. Looking at the results, it has been seen that oversampling methods give successful results in almost all cases. When all the outputs are examined, the XGBoost model, in which Tomek's Links method is applied, which is the most successful among all models. The outputs of the study are promising for the future. In the future, it is planned to use camera images together with log data to develop a more advanced model.

## REFERENCES

- [1] F. F. I. E. Council, "Supplement to authentication in an internet banking environment," pp. 206–222, 2011, [Online]. Accessed: August 15, 2023. [Online]. Available: <http://www.ffiec.gov/pdf/Auth-ITS-Final.pdf>
- [2] European Central Bank, "Recommendations for the security of internet payments," Technical Report, January 2013.
- [3] B. Alabi and A. David, "Framework for detection of fraud at point of sale on electronic commerce sites using logistic regression," *EAI Endorsed Transactions on Scalable Information Systems*, vol. 10, no. 2, p. e16, Nov. 2022.
- [4] C. Hines and A. Youssef, "Machine learning applied to rotating check fraud detection," in *2018 1st International Conference on Data Intelligence and Security (ICDIS)*, 2018, pp. 32–35.

- [5] G. R. Collins, "Safeguarding restaurants from point-of-sale fraud: an evaluation of a novel theft deterrent application using artificial intelligence," *Journal of Hotel & Business Management*, vol. 1, pp. 1–6, 2013.
- [6] K. Long, Y. Wu, and Y. Gui, "Anomaly detection of store cash register data based on improved LOF algorithm," *Appl. Math. (Irvine)*, vol. 09, no. 06, pp. 719–729, 2018.
- [7] Z. Xing, J. Pei, and E. Keogh, "A brief survey on sequence classification," *SIGKDD Explor. Newsl.*, vol. 12, no. 1, p. 40–48, nov 2010.
- [8] P. Gabbur, S. Pankanti, Q. Fan, and H. Trinh, "A pattern discovery approach to retail fraud detection," in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '11. New York, NY, USA: Association for Computing Machinery, 2011, p. 307–315.
- [9] F. Yoseph and M. Heikkilä, "A clustering approach for outliers detection in a big point-of-sales database," in *2019 International Conference on Machine Learning and Data Engineering (iCMLDE)*, 2019, pp. 65–71.
- [10] E. A. Lopez-Rojas, "Extending the retsim simulator for estimating the cost of fraud in the retail store domain," in *Proceedings of the European Modeling and Simulation Symposium, 2015*, 2015.
- [11] E. Lundin, H. Kvarnström, and E. Jonsson, "A synthetic fraud data generation methodology," in *Proceedings of the 4th International Conference on Information and Communications Security*, ser. ICICS '02. Berlin, Heidelberg: Springer-Verlag, 2002, p. 265–277.